

Exercise 4

Difference between JPA, Hibernate and Spring Data JPA

Feature	JPA	Hibernate	Spring Data JPA
Type	Specification	Framework (Implementation of JPA)	Framework built on top of JPA
Developed By	Oracle / Java EE	Hibernate Team	Spring Team
Purpose	Defines ORM standards	Implements JPA and provides ORM features	Simplifies database operations using JPA
SQL Writing	No	No	No
Boilerplate Code	Medium	Medium	Very Less
CRUD Support	Through implementation	Manual implementation	Built-in through JpaRepository
Dependency	Needs an implementation	Can work standalone or as JPA provider	Depends on JPA

Relationship

Application -> Spring Data JPA -> JPA (Specification) -> Hibernate (Implementation) -> MySQL Database

Example Comparison

Hibernate

```
Session session = sessionFactory.openSession();
```

```
Country country = session.get(Country.class, "IN");
```

You manage the session and transactions yourself.

Spring Data JPA

```
countryRepository.findById("IN");
```

That's it.

No session handling.

No SQL.

No implementation class.

Spring creates everything automatically.

Why Spring Data JPA is easier

Without Spring Data JPA, you would need to:

- Open a Hibernate Session
- Begin a Transaction
- Write HQL/Criteria or SQL
- Commit the Transaction
- Close the Session

With Spring Data JPA, you simply write:

```
countryRepository.findAll();
```

or

```
countryRepository.save(country);
```

Spring handles the rest.